

(12) **Patent Application Publication**
Heller

(43) **Pub. Date:** **Sep. 11, 2025**

Publication Classification

H04L 9/40 (2022.01)

G06F 21/72 (2013.01)

(52) **U.S. Cl.**
CPC **H04L 63/0428** (2013.01); **G06F 21/72**
(2013.01)

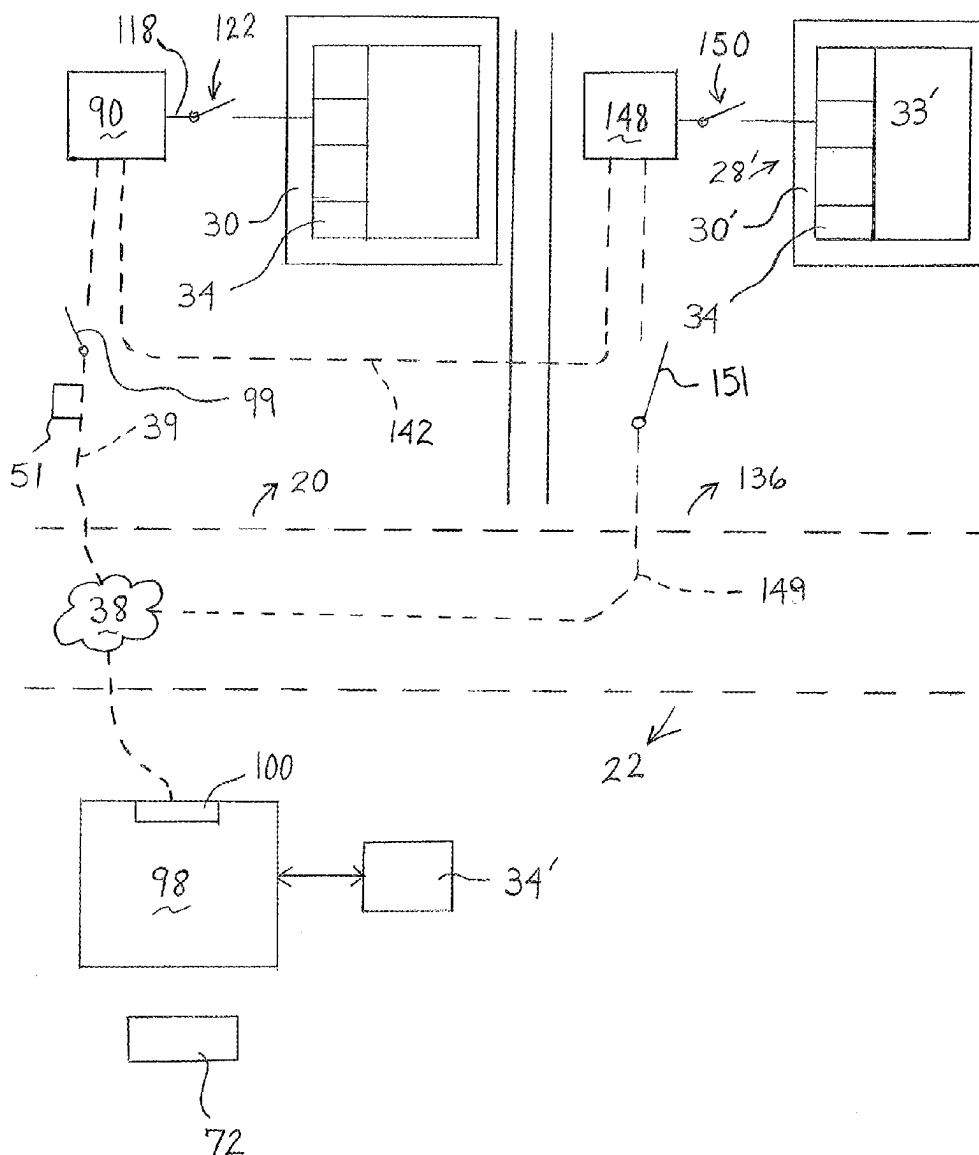
(57) **ABSTRACT**

(22) Filed: **Sep. 13, 2024**

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/457,532, filed on Dec. 3, 2021, now Pat. No. 12,118,104, which is a continuation of application No. 16/711,650, filed on Dec. 12, 2019, now Pat. No. 11,196,718.

A method of secure data transfer and storage using a storage device storing encrypted information. The method uses a host that stores and transfers encrypted sensitive information and a customer that desires the information to be securely stored. The customer chooses a unique encryption code to encrypt sensitive information and uploads the information to a host. Then the host transfers the information to the storage device. Connections between the host and customer are intermittent and done so the storage device remains isolated from networks outside of the host.



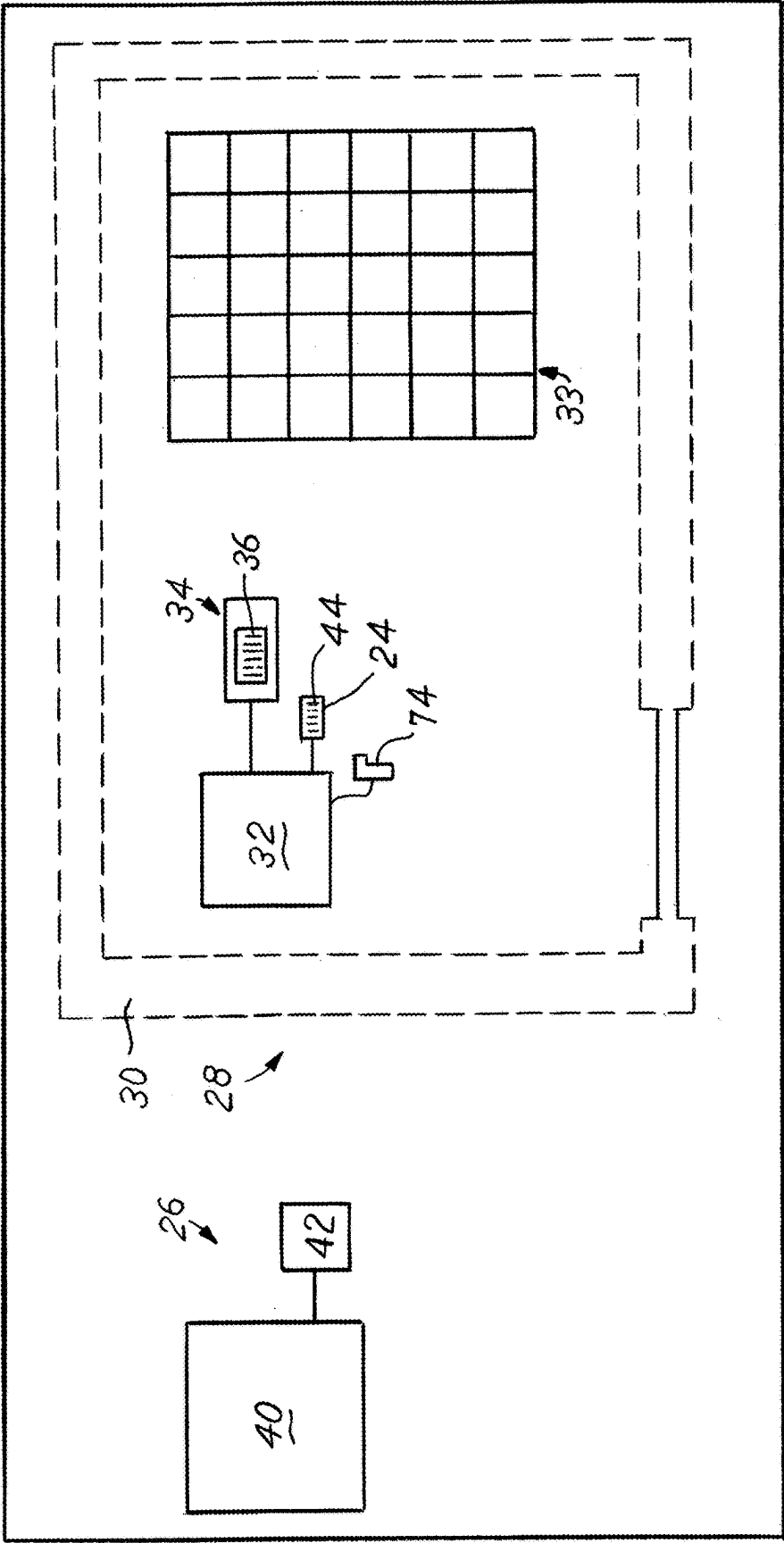
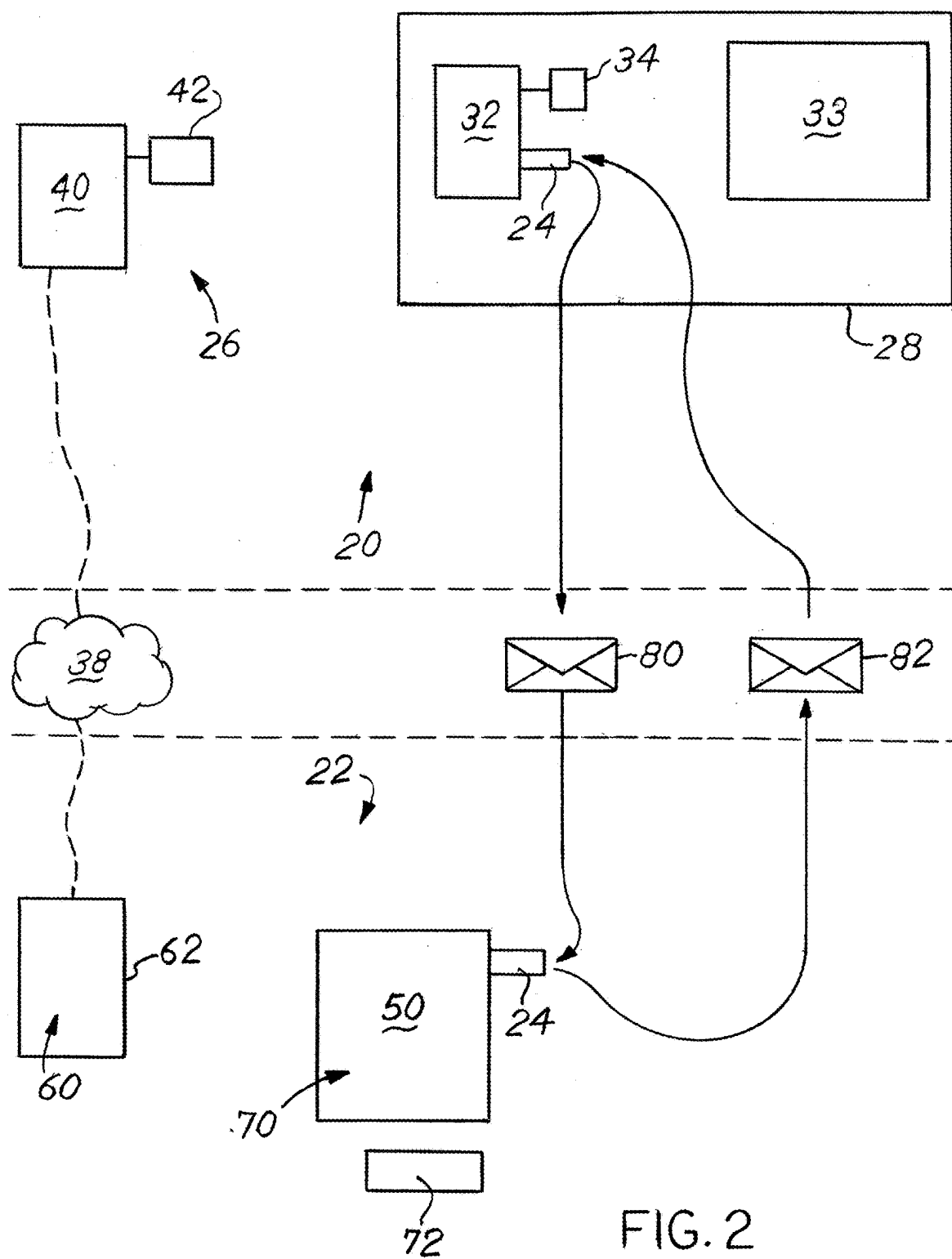


FIG. 1



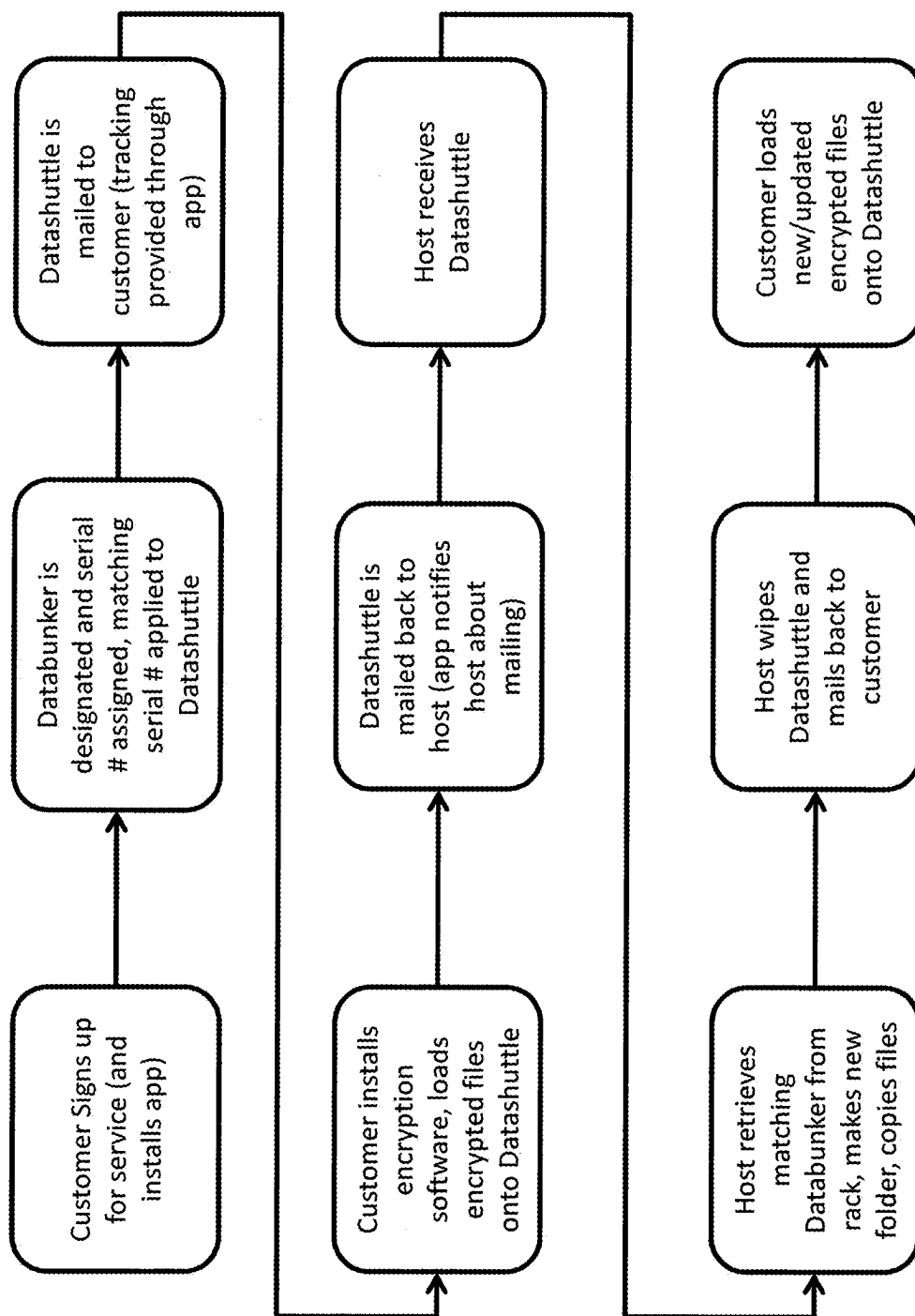


FIG. 3

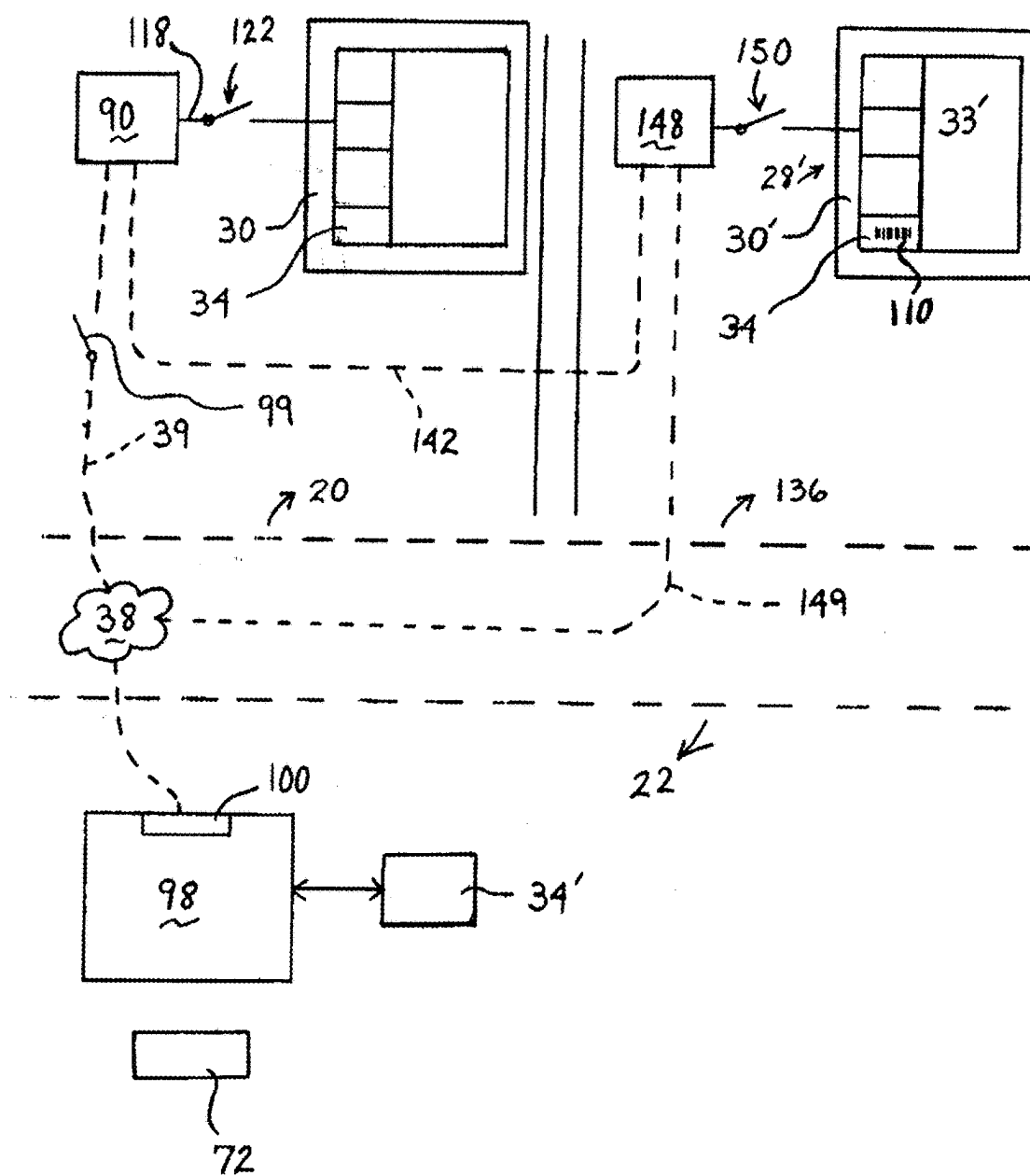


FIG. 4

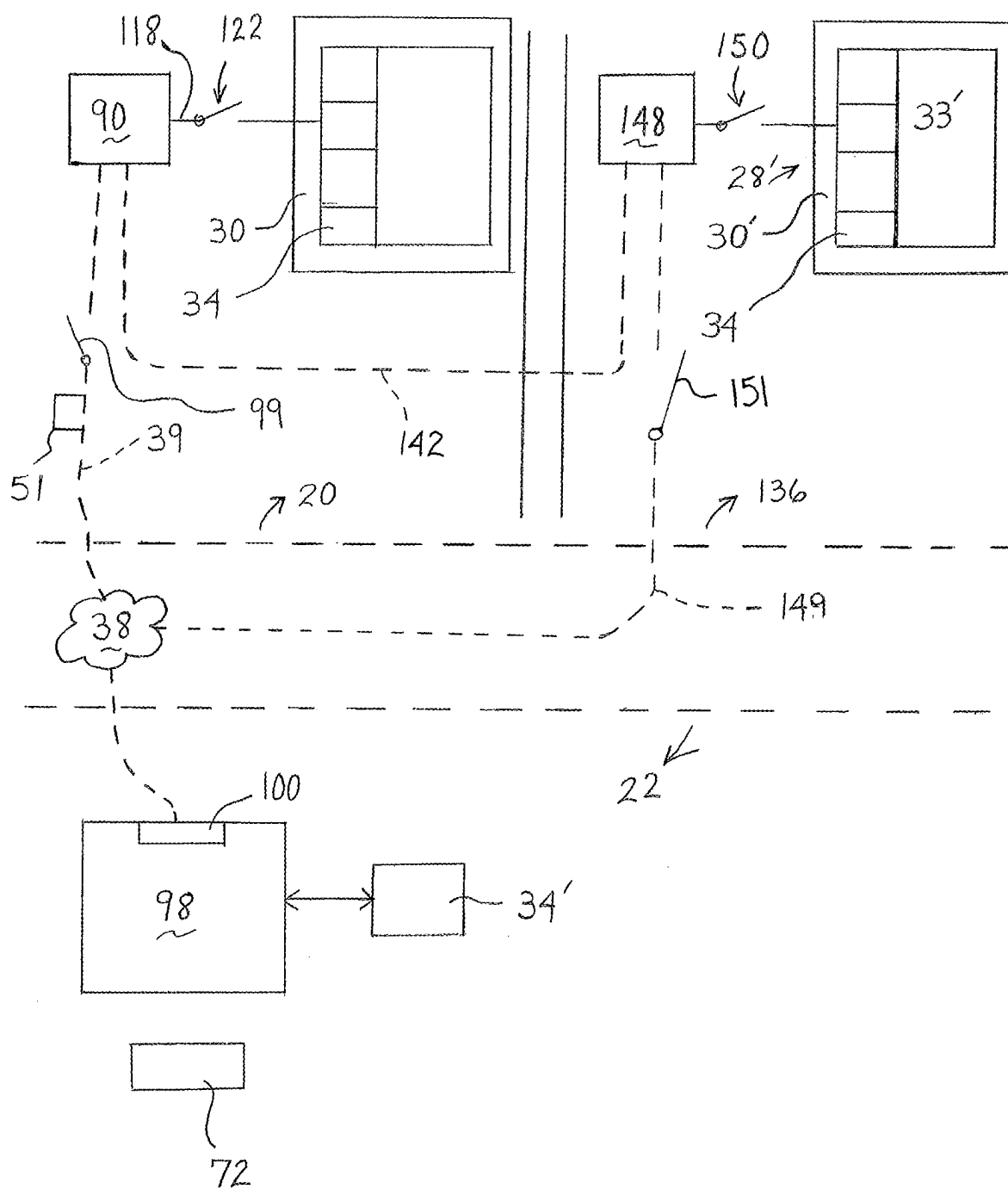


FIG. 5

METHOD OF SECURE DATA STORAGE AND TRANSFER

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation-in-Part and claims the benefit of priority from U.S. patent application Ser. No. 17/457,532, filed Dec. 3, 2021, which claims the benefit of U.S. Utility patent application Ser. No. 16/711,650, filed on Dec. 12, 2019, and the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] This present disclosure relates to information stored on computer disks, USB devices, and removable storage. There are multiple ways data can be destroyed, compromised, or damaged. Data can be destroyed through hardware failure, such as a hard drive crash or power surge, or through fire, flood, or mechanical force. Data stored on networked computers can also be compromised through hacking over the internet. It can also be compromised by careless behavior such as leaving a computer logged in and unattended, leaving an unsecured laptop out where it can get stolen, or losing a USB flash drive with sensitive information stored on it. Some systems and methods attempt to solve these problems, but often fall short. There are encryption algorithms to prevent a loose flash drive from being read, antivirus software to reduce the chances of a cyberattack, and enclosures to prevent destruction from fire, flood, or mechanical damage. Each system has individual vulnerabilities. An improved method is needed.

SUMMARY OF THE INVENTION

[0003] The present disclosure describes a method of using an air-gapped computer system in concert with ruggedized, encrypted on-site storage and portable encrypted storage. The method involves transfer of information from a customer's location to a secure location. The files are encrypted at the customer's location and sent over a secure connection. An intermittently gapped host computer is selectively isolated from any network outside of the host and can receive information from the customer over the secure connection. The intermittently gapped host computer may be connected to a storage device only when the intermittently gapped host computer is isolated from any network outside of the host. File transfer occurs between the on-site storage and removable storage only on the air-gapped computer system. The on-site storage retains previous versions of information.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A preferred embodiment of this invention has been chosen wherein:

[0005] FIG. 1 is a diagram of the system at the host;

[0006] FIG. 2 is an overall diagram of the systems at the host, customer, and the flow of information between them;

[0007] FIG. 3 is a flow diagram for when the customer sends sensitive information to the host;

[0008] FIG. 4 is a block diagram of how data is transferred when an internet connection is used with an intermittently gapped host computer; and

[0009] FIG. 5 is a block diagram of how data is transferred when two factor authentication is used.

DESCRIPTION OF THE PREFERRED EMBODIMENT

[0010] An apparatus and method of securely storing and updating information is described herein, involving a host 20, a customer 22, and the physical transportation of a removable device 24. The removable device 24 is made to withstand rough handling and may incorporate ruggedizing materials and manufacturing. For example, the removable device 24 may be integrated into a durable and/or resilient enclosure. The enclosure may include padding, other mechanical shock protection, and/or protection from moisture. The host 20 is the secured location where sensitive information is stored and archived. The customer 22 is located separately from the host 20 and has data that the customer desires to be securely backed up. Sensitive information is transferred between the host 20 and customer 22 completely offline, meaning the mail, hand-delivery, courier, or other method that is completely separate from any computer network. All file transfer and storage outside of the customer is encrypted and inaccessible without a customer-chosen unique encryption key 72. Any intercept of the removable device 24 by an unauthorized user will result in unreadable and inaccessible information without the encryption key 72.

[0011] The host 20 is a secure location and has a connected area 26 and an air-gapped area 28. While they are shown adjacent, the connected and air-gapped areas 26, 28 may be separate buildings, locations, or rooms. The connected area 26 is connected to the internet 38 but may have restricted access. The restricted access may include physically restricting access and/or restricting computer and network access. The air-gapped area 28 is completely separate from any outside communications and is surrounded by an "air-gap" 30, meaning there is no internet, intranet, or other network connectivity to any devices contained inside the air-gapped area 28. The air-gapped area 28 is protected from outside unauthorized access at the host 20. The protection may be in the form of lock and key, key cards, physical keys, access codes, RFID tags, or other systems to restrict physical access to the air-gapped area 28. Located inside the air-gapped area 28 is an array 33 of armored storage units 34, each having their own barcoded serial number 36 or other unique identification. The armored storage units 34 described herein are metal enclosures with extreme heat-resistant insulation and components, but it is contemplated that other types of storage devices are used, such as removable hard drives or other storage media. It is contemplated that the armored storage units 34 are stored in fire resistant enclosures. The air-gapped area 28 may be supplied by an independent power source, backup generator, or otherwise be connected to an off-grid supply of power. The air-gapped area 28 further contains an air-gapped host computer 32 that may be located adjacent the array 33 or in a separate location. The air-gapped host computer 32 is completely standalone from any computer network, including any computer network at the host. Further, the air-gapped host computer 32 may contain a custom operating system, software, or firmware that is used for copying and updating sensitive information between the storage unit 34 and removable storage device 24. The storage units 34 are normally in a disconnected state and reside in the storage array 33, where no electricity is present and no communication to the storage unit 34 can occur. The storage units 34 are rugged and are protected from high heat, mechanical damage, liquid intrusion, and

electrical events. Each storage unit **34** has a rigid metal shell with a tight-fitting lid. Inside the metal shell is high-temperature insulation that surrounds a storage unit. The storage unit may include phase-change heat absorbers that surround a solid-state storage device. Sacrificial wiring connects the solid-state storage device through a small gap in the rigid metal shell. In the event that there is a fire, flood, earthquake, collapse, or other damaging event inside the air-gapped area **28**, each storage unit **34** is individually protected. This is in contrast to a server or drive array that relies on protection supplied by the room or area where it is installed. In a server or drive array, a flood or fire could easily cause destruction of all data and hardware.

[0012] The connected area **26** has an internet-connected host computer **40**. For simplicity, the internet-connected host computer **40** is shown and described as a single computer, but the internet-connected host computer **40** can be a network or series of computers that may be completely located at the host or hosted off-site. There is no communication between the internet-connected host computer **40** and the air-gapped host computer **32**. The internet-connected host computer **40** has a barcode printer **42**, which is located at the host **20**. The internet-connected host computer **40** may have a database that stores customer information, such as the mailing address, billing information, file transfer log, serial number/ID of their storage unit **34**.

[0013] The customer has its own computer **50** or computer network that uses sensitive information. The sensitive information may be stored on a network drive, attached storage, or on the computer(s) directly. The customer's computer **50** may be connected to the internet **38** or be on an air-gapped network. Sensitive information can include trade secrets, banking information, programs, personnel records, accounts and passwords, databases, or any type of electronic files or information that may be important to the customer. The term "sensitive information" is used as a definition for the files that are stored and transferred between the host and customer and is not limiting to the types of files that are stored, encrypted, and transferred. While the files are commonly sensitive or contain sensitive information, the files can also be ordinary and/or publicly-available.

[0014] An app **60** may be provided by the host that the customer installs on a device **62** that is connected to the internet **38**. This is commonly a smartphone, tablet, or other mobile device. The app **60** communicates with the connected host computer **40** through the internet **38** and is used to initiate the offline transfer of sensitive information. The app **60** also is used to track any shipment/delivery of the removable storage device **24**. Alternatively, the customer **22** may contact the host computer **40** through the internet via web portal, phone call, or email. The customer installs encryption software **70** and chooses a unique encryption key **72** that is not shared with anyone at the host **20**. Without the encryption software **70** and key **72**, the sensitive information is completely inaccessible, even by the host **20**. If the customer **22** loses the encryption key **72**, the data is inaccessible.

[0015] The air-gapped host computer **32** may include a barcode reader **74** to document and ensure that the barcodes **36**, **44** match before any file transfer. Because the files are not readable by the host, ensuring matching serial numbers prevents errors and loss of information.

[0016] There are different scenarios when transferring sensitive information. These include, but are not limited to,

the following: initial load, updating, and restoration. In the initial load, the sensitive information is only at the customer's computer **50** and no sensitive information is present at the host **20**. When updating, the sensitive information at the customer's computer **50** has been updated or changed, and the customer desires the updated sensitive information to be stored at the host. During restoration, the sensitive information has been compromised or destroyed and the sensitive information stored at the host **20** is the only trusted copy.

[0017] Alternatively, the customer **22** desires to move it to a new computer or network. In all scenarios, the sensitive information never travels between the host **20** and customer **22** via a computer network or the internet **38**. The files are only transferred by physical delivery.

[0018] For the initial load, the customer **22** contacts the host for a blank removable device **24**. The connected host computer **40** prints out a unique barcode **44** that is tied to that customer and a designated storage unit **34**. The unique barcode **44** matches on the removable device **24** and storage unit **34**. The barcode **44** is applied to the blank removable device **24** and then the host **20** places the blank device **24** in the mail **80** to the customer. The mail **80** is tracked by the host **20** to ensure safe delivery to the customer **22**. The customer **22** installs encryption software **70** and chooses an encryption key **72**. If the encryption key **72** is lost, any encrypted information is inaccessible. Upon receipt of the removable device **24**, the customer **22** connects the blank removable device **24** to the customer computer **50** and enters the encryption key **72**. The customer **22** encrypts and loads all sensitive information onto the removable device **24**. This takes the sensitive information and transforms it to encrypted sensitive information as it is being loaded onto the removable device **24**. Upon completion, the customer **22** then mails **82** the removable device **24** to the host **20**. The host **20** tracks the package, either through the app **60** or being otherwise notified by the customer **22**. The sensitive information on the removable device **24** is encrypted. When the removable device **24** arrives at the host **20**, the barcode **44** is scanned at the internet-connected host computer **40** to tie it to the customer's account. The removable device **24** is never connected to the internet-connected host computer **40**, only the air-gapped host computer **32**. The host **20** identifies the correct armored storage unit **34** in the air-gapped area **28**. The armored storage unit **34** and removable device **24** are connected to the air-gapped computer **32** and the encrypted files are copied from the removable device **24** to the armored storage unit **34**. After file transfer is complete, the armored storage unit **34** is disconnected and placed back in the array **33**. The host **20** cannot access files on either device **24**, **34** due to the encryption. The removable device **24** is then wiped clean of information and mailed **80** back to the customer **22**. The customer **22** may be notified through the app **60** that the transfer is complete and the blank removable device **24** is on its way back to the customer **22**.

[0019] To update the sensitive information, the customer **22** takes the removable device **24** and loads encrypted sensitive information onto the removable device **24**. The customer **22** then notifies the host **20** through the app **60** and mails **82** the removable device **24** to the host **20**. When the host **20** receives the removable device **24**, the encrypted files are copied from the removable device **24** to the armored storage unit **34**. A new folder is added to the armored storage unit **34** with the updated information. Each time the sensitive information is updated, a new folder is created to allow

previous versions of sensitive information be retrievable by the customer in the event one version of the sensitive information is corrupted or otherwise compromised. It is contemplated that alternate methods of saving and retrieving versions of sensitive information are used.

[0020] To restore sensitive information, the customer 22 contacts the host 20 through the app 60. The host 20 may either load a new removable device 24 with sensitive information or mail 80 the armored storage unit 34 back to the customer 22. As with other mailing or shipping, the progress is tracked through the app 60.

[0021] In some instances, sensitive information may be transferred between the host 20 and the customer 22 through a secure online connection. This is shown in FIG. 4. In this case, the host 20 is located in a secure location and has an intermittently connected host computer 90 located with the host 20. The intermittently connected host computer 90 is connected to the internet 38 via a secure connection 39. The secure connection 39 may include SSL. The secure connection 39 to the internet 38 is between a device 98 located with the customer 22 and the intermittently connected host computer 90. The information leaving the device 98 is encrypted 100 before it leaves the customer 22. The device 98 at the customer 22 may be a computer or local cloud and may even be a network of computers. Once the information is encrypted 100, that information may only be viewed by using an encryption key 72 that the customer 22 retains. In addition to the information being encrypted before it leaves the customer 22, the information may be encrypted at a server on the internet 38 between the device 98 located with the customer 22 and the intermittently connected host computer 90. Thus, any information that may be potentially intercepted on the internet 38 would be unreadable without the key 72 and that is still true if the information were decrypted during its time on the internet 38 because such decryption would only be for the second layer of encryption. The device 98 may include special software that strictly limits the amount of time information is sent via the internet 38. For instance, the software on the device 98 may intermittently transmit information through the internet 38 by only activating a connection to the internet 38 on a predetermined time interval. The software may be set up to send information on the internet 38 once every five minutes in a burst of packets. Such transmissions periodically limit the amount of time the information spends in transit on the internet 38 and minimizes the time it may be intercepted. The user of the device 98 may select the interval of each transmission of information to a longer or shorter time. Such a choice for the predetermined time interval before sending information via the internet 38 would depend on the user and a shorter time interval before sending information would limit the amount of time information is not synchronized with the intermittently host computer 90. In addition to the user limiting the amount of time information is transmitted over the internet 38 at the device 98, the intermittently connected host computer 90 may do so as well. Before connecting to the intermittently connected host computer 90, a login is required to make that connection.

[0022] The login may be monitored through a firewall 51 between the device 98 and the intermittently connected host computer 90. To add a further layer of security, the intermittently connected host computer 90 may have a switch 99 to affirmatively break its connection to the internet 38 and provide the intermittently connected host computer 90 an

air-gapped connection. The firewall 51 is connected to the switch 99 such that the firewall 51 can prevent the switch 99 from closing. When the device 98 attempts to login to the intermittently connected host computer 90 to make a secure connection, the host 20 sends a unique identifier code 55 to the customer 22. The host 20 can send this unique identifier code 55 to the customer 22 in number of different ways, including through the internet 38. The host 20 can send the unique identifier code 55 to the same device which the customer 22 is attempting to securely connect to with the intermittently connected host computer 90. This unique identifier code 55 can be sent to the customer 22 through a text or email to the same device 98 or a different device. The host 20 also ensures that the firewall 51 has a copy of the unique identifier code 55 sent to the customer 22. The login directs the customer 22 to enter a code. The firewall 51 compares the code received from the customer 22 to the code received from the host 20. If the code entered by the customer 22 into the input box does not match the unique identifier code 55, the firewall 51 does not allow the switch 99 to close and the customer 22 cannot connect to the intermittently connected host computer 90. When the customer 22 enters the unique identifier code 55 into the input box, the firewall 51 does not prevent the switch 99 from closing. Thus, the firewall 51, along with the login, ensures that the connection between the host 20 and the customer 22 is a secure connection confirmed by multiple forms of identification. The customer 22 must use the correct login and must match the unique identifier code 55 in the input box before the intermittently connected host computer 90 can connect to the device 98 through the internet 38. The firewall-protected switch 99 is never closed at the same time as the switch 122 between the intermittently connected host computer 90 and the armored storage unit 34. This ensures that the air-gapped area 28 remains truly air gapped and secure. The firewall 51 can never be used to over-ride the other safeguards so that the switch 99 and switch 122 are closed at the same time. The firewall 51 ensures that the switch 99 remains open and the intermittently connected host computer 90 remains disconnected from the device 98 through the internet 38. In this way, the firewall 51 ensures that the intermittently connected host computer 90 only connects to the customer's device 98 and not some other device. Because the firewall 51 and switch 99 prevent the intermittently connected host computer 90 from sending or receiving packets of information through the internet 38 to any device other than the customer's device 98, the connection is secure.

[0023] The intermittently connected host computer 90 also has the capacity to monitor activity from a customer 22 connecting to it or attempting to connect to it. In the absence of activity, the intermittently connected host computer 90 could time the customer 22 out after a predetermined time thereby breaking the connection to the internet 38 to minimize the time the intermittently connected host computer 90 and the device 98 are connected.

[0024] The intermittently connected host computer 90 retains a serial number 110, or other unique identification, that is provided to the customer 22 and associated with an armored storage unit 34. The associated serial number 110 of the armored storage unit 34 is also associated with a particular customer 22 and is stored in an array 33 of armored storage units 34. As described, the array 33 of armored storage units 34 are stored in an air-gapped area 28. Each

armored storage unit 34 is an extremely durable unit that has the capacity to survive many types of disasters including fire, flood, or most types of damaging event that would destroy a less durable storage device.

[0025] The intermittently connected host computer 90 has an intermittent connection 118 to the customer 22 and also has an intermittent connection 118 to storage units 34 that correspond to particular customers 22. An intermittent connection 118 is made to each storage unit 34 within the array 33. The intermittent connection 118 is gapped via the use of a switch 122 or other gapping device that does not leave the storage unit 34 connected while the intermittently connected host computer 90 is connected to the customer 22 via the internet 38. The switch 122 may be a relay or other type of switch that is electrically controlled to physically break the connection 118 between the intermittently connected host computer 90 and the storage units 34 within array 33. Each armored storage unit 34 must be connected to the intermittently connected host computer 90 through a switch 118 and although only one switch 118 is shown, there may be many switches connecting the intermittently connected host computer 90 to each storage unit 34 within the array 33. In this manner, there is a physical break between each storage unit 34 and the world outside of the air-gapped area 28. Generally, for a fully gapped air-gapped area 28, the switches 122 to any one of the storage units 34 in the air-gapped area will not be connected while the intermittently connected host computer 90 is connected to the internet 38. In other words, if the connection to the internet 38 is viewed as a switch 122 that is closed when the intermittently connected host computer 90 is connected to the internet 38, the switch 122 is open so that the air-gapped area 28 remains truly air gapped. More simply stated, the switch 122 will never be closed when the intermittently connected host computer 90 is connected to the internet 38. The intermittently connected host computer 90 may be configured to store information from particular customers 22 it receives throughout a time period, then, for a select timeframe, deny all connections to the internet 38 so that switches 122 to particular storage units 34 corresponding to customers 22 who sent information may be transferred through the connection 118 into the air-gapped area 28. In this manner, the air-gapped area 28 may be truly air-gapped because no computer or device connected to a storage unit 34 is connected to the internet 38 or any other network when information is being transferred to the storage unit 34. As such, although the intermittently connected host computer 90 may at times be connected to the internet 38, it is not connected to the internet 38 when it is connected to a storage unit 34. Therefore, the intermittently gapped host computer is selectively isolated from any network to facilitate secure transfer of information to the storage units 34.

[0026] To add a further layer of security, the intermittently connected host computer 90 may have a switch 99 to affirmatively break its connection to the internet 38 and provide the intermittently connected host computer 90 an air-gapped connection. The storage units 34 are always isolated from any network. The delay that the intermittently connected host computer 90 has while it holds information before being transferred to a corresponding storage unit 34 may be set to a predetermined time. If all information received via the internet 38 at the intermittently connected host computer 90 that is destined for a particular storage unit 34 is updated via connection 118 after the normal business

hours of the host 20 during the night, a customer 22 seeking to retrieve his particular storage unit 34 the next morning can be assured that it will contain the information sent on the previous day. The general purpose of the network shown in FIG. 4 is to have a nearly continuously updated version of the device 98 located at the customer 22 to be replicated on the storage unit 34. In addition to the storage unit 34 located at the host 20, the customer 22 may also have a locally located storage unit 34' that is the same as that in the array 33 at the host 20. Such a locally located storage unit 34' is continuously synchronized with the device 98.

[0027] In addition to having a host 20, there may be a second host 136 located remotely from the host 20. The second host 136 may be a mirrored host having all of the same information stored within its array 33' as in the array of the host 20. The second host 136 may be connected through a direct connection 142 to the host 20. This direct connection 142 may be separate from the internet 38 provided the second host 136 is in a location that provides the ability to have a direct connection 142 apart from the internet 38. In other instances, the direct connection 142 to the second host 136 may be through the internet 38. The second host 136 also has an intermittently connected computer 148 that connects to the array 33' via a switch 150. Switch 151 may be used to selectively disconnect the intermittently connected computer 148 from the internet 38. In this configuration when switch 99 or 151 are closed, neither switch 122 nor 150 may be closed. This will maintain the air-gap and ensure storage units 34 are never connected to the internet 38. Thus, the second host 136 may have an air gap 30' surrounding air-gapped area 28' that functions as the air-gapped area 28 at the host 20.

[0028] It is understood that while certain aspects of the disclosed subject matter have been shown and described, the disclosed subject matter is not limited thereto and encompasses various other embodiments and aspects. No specific limitation with respect to the specific embodiments disclosed herein is intended or should be inferred. Modifications may be made to the disclosed subject matter as set forth in the following claims.

What is claimed is:

1. A method of secure data transfer and storage between a host and a customer, said method comprising:
 - said host providing an intermittently gapped host computer in a secure location, said intermittently gapped host computer selectively isolated from any network;
 - said host providing a storage unit in said secure location, said storage unit isolated from any network outside of said host;
 - providing a secure connection between said intermittently gapped host computer and a customer computer;
 - serializing said storage unit;
 - said customer attempting to make said secure connection to said intermittently gapped host computer and said host generating a code for said customer and sending said code to said customer;
 - said customer communicating said code to said host and upon said customer matching said code from said host, said host allowing said secure connection;
 - said customer encrypting and transferring sensitive information from said customer computer to said intermittently gapped host computer via said secure connection;

disconnecting said intermittently gapped host computer from said secure connection; and
 said host transferring said encrypted sensitive information from said intermittently gapped host computer to said storage unit having a serial number corresponding to said customer while said intermittently gapped host computer is disconnected from said secure connection and any network outside of said host.

2. The method of secure data transfer in claim 1, wherein said secure connection is over an internet.

3. The method of secure data transfer in claim 1, wherein said storage unit is in an array including other storage units and said intermittently connected host computer is selectively connected to said storage unit.

4. The method of secure data transfer in claim 3, wherein said array is in an air-gapped area.

5. The method of claim 1, wherein a switch is located in a connection between said intermittently connected host computer and said storage unit, said switch open when said intermittently connected host computer is connected to said secure connection and said switch being closed only when said intermittently connected host computer is not connected to said secure connection.

6. The method of claim 5, wherein said secure connection to said intermittently connected host computer may be connected and disconnected with a switch.

7. A method of secure data transfer and storage for sensitive information comprising the steps of:
 providing an intermittently gapped host computer in a secure location, said intermittently gapped host computer selectively connectable to an internet connection;
 said host providing a storage unit in said secure location, said storage unit isolated from any network outside of said host;
 a customer computer being connectable to said intermittently gapped host computer via said internet connection;
 assigning a unique serial number to said storage unit;
 encrypting said sensitive information on said customer computer;

said customer attempting to make said secure connection to said intermittently gapped host computer and said host generating a code for said customer and sending said code to said customer;

said customer communicating said code to said host and upon said customer matching said code from said host, said host allowing said secure connection;

connecting said customer computer to said intermittently gapped host computer via said internet connection and transferring said encrypted sensitive information to said intermittently gapped host computer;

maintaining a database storing customer information and said unique serial number corresponding to said storage unit;

disconnecting said intermittently gapped host computer from said internet connection and transferring said sensitive information from said intermittently gapped host computer to said storage unit only when said intermittently gapped host computer is disconnected from said internet connection.

8. The method of claim 7, wherein said intermittently connected host computer is selectively connected to said internet connection with a switch, disconnecting said switch providing an air-gap when said switch is open.

9. The method of claim 7, wherein said storage unit is in an array including other storage units and said intermittently connected host computer is selectively connected to said storage unit.

10. The method of claim 9, wherein said array is in an air-gapped area.

11. The method of claim 7, wherein a switch is located in a connection between said intermittently connected host computer and said storage unit, said switch open when said intermittently connected host computer is connected to said internet connection and said switch being closed only when said intermittently connected host computer is not connected to said internet connection.

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